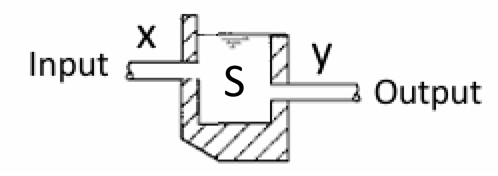
# What is memory?

In its simplest definition, “**Memory**” is something which is able to **store the previous states**.

In other words, “**Memory**” enables the existence of “**State Space**”.

In order to understand what really the **state space** is, we could look at the following water tank:




The Output Flow **Y** is dependent on the Input  Flow **X**, as well as the Capacity of Storage **S**, which is actually able to define the **State Space**.

$$\mathbf{Y} = F(\mathbf{X}, \mathbf{S})$$

In such case, we could loosely define the Water Tank as the **Memory of the Flow** of water.

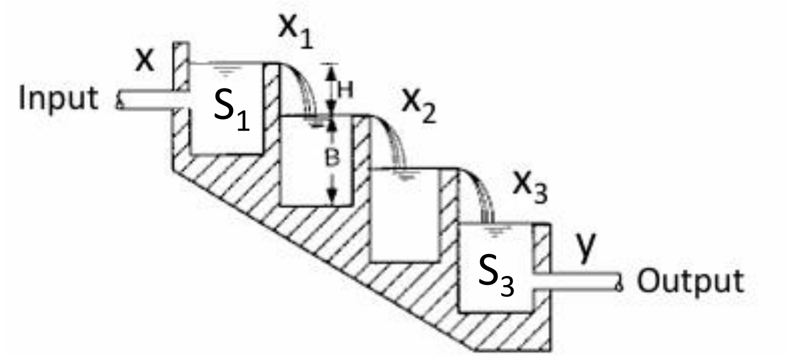
Since the output **Y** is dependent on both **X** and **S**, let us define a **Deterministic Causality**, such that:

If **X**, then **Y**, given **S**

If we know the State **S** and Input **X**, then we can **definitely determine** the Output **Y**.

# State Space

Now, let us look at a more interesting State Space, in which we have Cascading Storage.




In such case, we can define the "**Chain of Deterministic Causality**", such that:

If **X**, then **$X_1$**, given **$S_1$**

If **$X_1$**, then **$X_2$**, given **$S_2$**

…

If **$X_n$**, then **Y**, given **$S_n$**

Thus, the State Space $\{S_1, \ldots, S_n\}$ defines the **Deterministic State Space**, which can be "**Inference**" or "**Reason**" the chain of causality from **X** to **Y**.

# Deterministic Causality

Given a State Space {$\mathbf{S}_1, \ldots, \mathbf{S}_n$ } (Memory), the Deterministic Causality is defined in such a way that:

$$\text{If } \mathbf{X}\text{, then } \mathbf{Y}\text{, given } \{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$$

Inversely:

$$\text{If } \mathbf{Y}\text{, then } \mathbf{X}\text{, given } \{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$$

As we can see, the Memory or State Space is something which can connect the past events to the present events, through **Causality**.

More importantly, it is such Deterministic Causality that defines the "**Rational Determinism**", or rather "**Rationality**", which is defined as **Aristotelian Logic**.

$$\text{If } \mathbf{X}\text{, then } \mathbf{Y}\text{, given } \{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$$

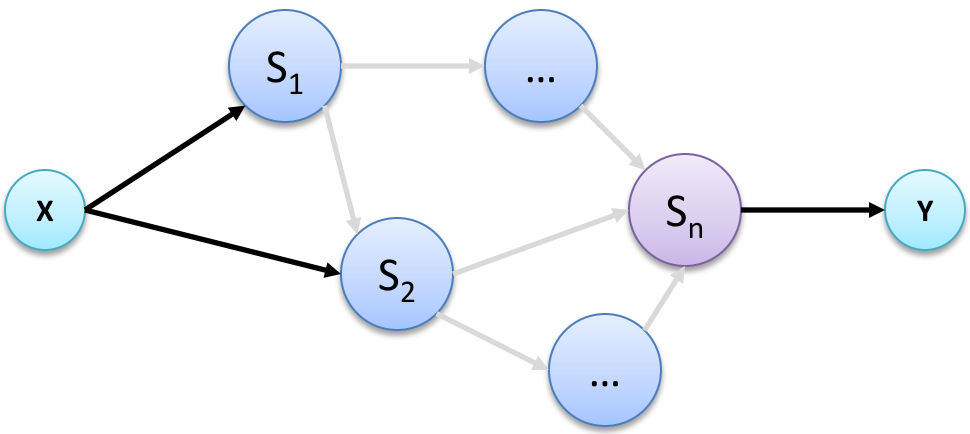
Given State Space {$\mathbf{S}_1, \ldots, \mathbf{S}_n$ }, if X happens, then Y must definitely happen, without exception.

As we can see Deterministic Causality has no room for "**Uncertainty**".

# Deterministic Reasoning

If **X**, then what is **Y**, given {$\mathbf{S}_1$, …, $\mathbf{S}_n$ }

With Deterministic Causality, **Deterministic Reasoning** is possible. What is Deterministic Reasoning is reaching the **Definite Output State**, given **State Space** and **Input State**.




Given Yangon city is 60 km away from Bago City (**S**), and if we are driving with a speed of 60 km per hour (**X**), then we can travel from Yangon to Bago in one hour (**Y**).

As we can see, **Deterministic Reasoning** will always lead to **Definite Output State**, which, however, leads to **Optimistic Reasoning**, which does not take account for any **Uncertainty**.

It is why **Deterministic Rationalism** always results in **Idealism**, which has no Dubious States.

# Non Deterministic Causality

However, in Reality, it is not always possible to reach a Definite Output **Y** from an Input **X**, given a state space {$\mathbf{S}_1$, …, $\mathbf{S}_n$ }.

Sometimes, more than an Output **Y** is possible from an Input **X**, given a state space {$\mathbf{S}_1$, …, $\mathbf{S}_n$ }.

Thus, **Non Deterministic Causality** is defined such that:

If **X**, then { $\mathbf{Y_1}$, …, $\mathbf{Y_m}$} given { $\mathbf{S}_1$, …, $\mathbf{S}_n$ }

In other words, given a State Space { $\mathbf{S}_1$, …, $\mathbf{S}_n$ }, an Input **X** could probably result in many possible Outputs { $\mathbf{Y_1}$, …, $\mathbf{Y_m}$}.

However, formally, it is really quite difficult to define **Non Deterministic Causality**, which is why **Stochastic Causality** is defined, along with **Probability**.

Actually, **Non Deterministic Causality** is more aligned with Reality, which results in **Realism**, which allows Dubious States.

# Stochastic Causality

**Stochastic Causality**, with **Probability**, is defined such that:

$$\text{If } P(\mathbf{X}), \text{ then } P(\mathbf{Y}), \text{ given } \{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$$

Given a State Space $\{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$, an Input **X**, with a Probability of P(**X**), could result in Output Y, with a Probability of P(**Y**).

Actually, we can define **Stochastic Causality** a bit differently, with **Bayes Theorem**, such that:

$$P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S})$$

However, with the assumption of **Conditional Independence**:

$$P(\mathbf{Y} \mid \mathbf{X},\mathbf{S}) = P(\mathbf{Y} \mid \mathbf{S})$$

Then, $P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S})$ varies with $P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S})$ as $P(\mathbf{X} \mid \mathbf{S})$ will be constant.

$$P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}) = P(\mathbf{Y} \mid \mathbf{S})\, P(\mathbf{X} \mid \mathbf{S})$$

$$P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}) = P(\mathbf{Y} \mid \mathbf{X},\mathbf{S})\, P(\mathbf{X} \mid \mathbf{S})$$

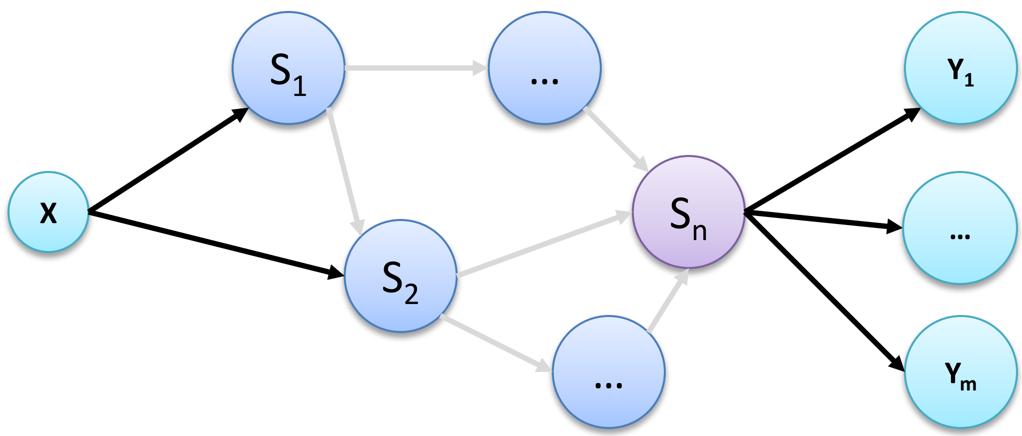
$$P(\mathbf{Y} \mid \mathbf{X},\mathbf{S}) = P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}) \,/\, P(\mathbf{X} \mid \mathbf{S})$$

$$P(\mathbf{Y} \mid \mathbf{X},\mathbf{S}) \;\alpha\; P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}), \;\; \text{if } P(\mathbf{X} \mid \mathbf{S}) = \text{Constant}$$

# Stochastic Reasoning

If P(**X**), what is P(**Y**), given {$\mathbf{S}_1$, …, $\mathbf{S}_n$ }

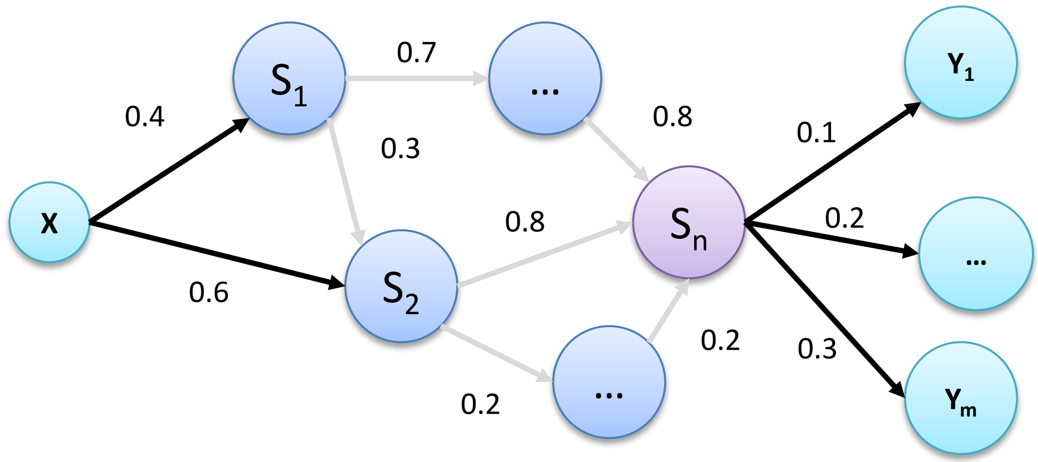
With Stochastic Causality, **Stochastic Reasoning** is reaching the **Probable Output States**, given the Probability of **State Space** and **Input State**.




Given Yangon city is 5 km away from the sea (**S**), and if we are in Monsoon Season (**X**), then what is the probability of rain in Yangon tomorrow (**Y**).

# Bayesian Network

Bayesian Network is an **Acyclic Network** of the Chain of **Bayesian Probability**, which is generally used in **Stochastic Reasoning**, or more simply **Bayesian Inference**.



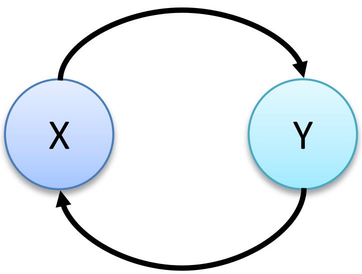

However, Bayesian Network also assumes **Conditional Independence**, which in turn forbids the Existence of **Loops** or **Cycles** in the network.

Therefore, Bayesian Network must be an **Acyclic Network** of **Chain of Conditional Probability**, without any Cyclic Dependency.

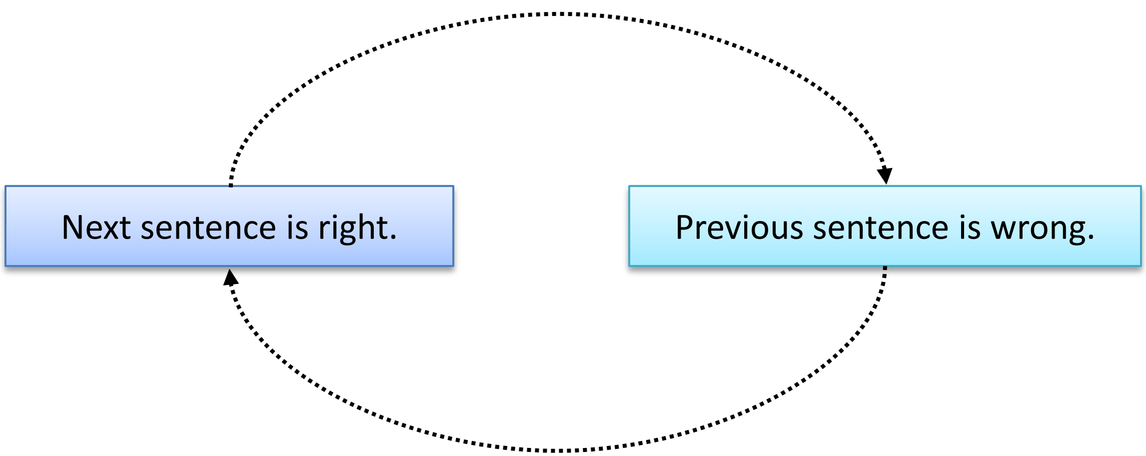
Reasoning , either Stochastic or Deterministic, is not perfect in that it cannot allow **Cyclic Inference**, which is interesting because most **Asian Philosophies** are based on Cycles.

# Cyclic Dependency

**Cyclic Dependency**, or rather **Cyclic Inference** is not generally allowed in any form of Reasoning, either Stochastic or Deterministic, as it could never reach **Rational Conclusion**.




For example, what is the logical conclusion of which one is correct?



# Stochastic Learning

**Reasoning**, either Deterministic or Stochastic, is reaching the Output **Y** from Input **X**, given the State Space **S**.

$$\text{If } P(\mathbf{X}), \text{ what is } P(\mathbf{Y}), \text{ given } \{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$$

Then, **Learning**, on the other hand, is:

$$\text{If } P(\mathbf{X}), \text{ what is } \{\mathbf{S}_1, \ldots, \mathbf{S}_n\}, \text{ given } P(\mathbf{Y})$$

It turns out that Learning, especially **Stochastic Learning**, is far more difficult than **Stochastic Reasoning**.

More precisely, Stochastic Learning is updating the State Space $\{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$, based on Input **X** and Output **Y**, which is extremely hard due to the various possibilities of states.

Actually, **Stochastic Learning** is a **NP Hard** Problem, which is why Bayesian Network is mainly used for **Bayesian Inference**, rather than **Bayesian Learning**.

Later, Stochastic Learning is made possible by a Network of Associative Neurons, which is now known as **Artificial Neural Network** (ANN).

# Gaussian Process

Before looking into Stochastic Learning, we would look into **Gaussian Process**, in which **Gaussian Model** is utilized for **Bayesian Inference**.

Suppose we have a Set of Inputs X = { $x_1$ , … , $x_n$}, and corresponding Outputs Y = { $y_1$ , … , $y_n$}.

Then we can implicitly define a function Y = f(X) + ∈, in which f(X) is defined as Gaussian Process with zero mean (μ), such that $\mathcal{N}$( μ = 0, $K_{xx}$), where $K_{xx}$ is a **Covariance Matrix** or **Kernel**, and ∈ is random variable.

We can actually define the Inputs and Outputs as a Joint Distribution P(X, Y), which is a **Multivariate Gaussian Distribution**.

$$P(X, Y) = \mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}K_{xx} & K_{xy}\\K_{yx} & K_{yy}+\sigma^2\end{bmatrix}\right)$$

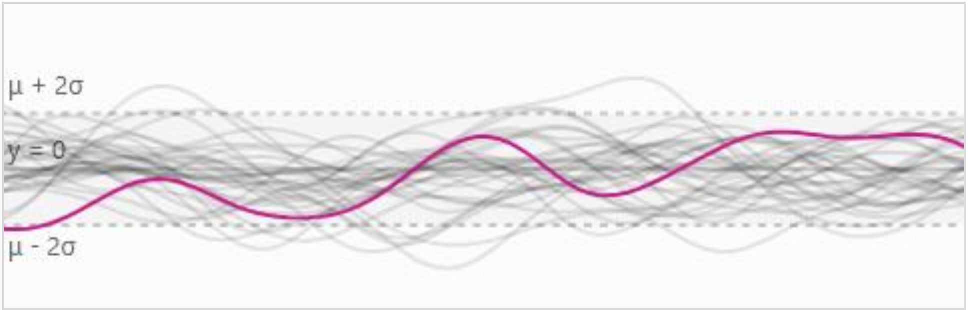
# Gaussian Process

Now, we would like to Inference new Output { y' }, given by new Input { x'}.

According to **Bayesian Inference**, **Posterior Distribution** P(Y | X) is estimated from **Prior Distribution** P(X).

In our case, our **Prior Distribution** can be defined as:

$$Y = f(X) + \in \; = \mathcal{N}(\mu = 0, K_{xx}) + \in$$



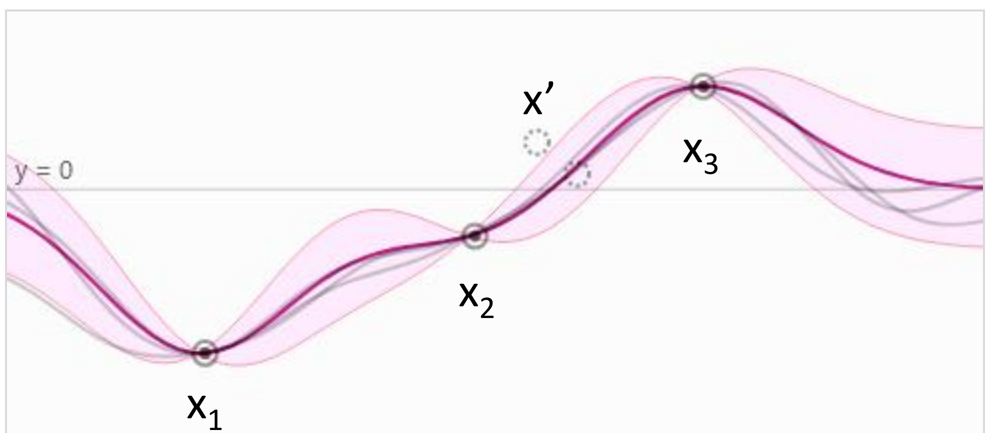

Now, we are going to estimate the Posterior Distribution from this Prior Distribution, from the Set of Inputs and Outputs { $x_1$ , … , $x_n$} and { $y_1$ , … , $y_n$} for a new Input { x' }.

# Gaussian Process

Since we have Set of Inputs and Outputs { $x_1$ , … , $x_n$} and { $y_1$ , … , $y_n$}, and we want to predict the value y’ given x’, then:

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \\ f(x') \end{bmatrix} = \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K_{xx} + \sigma^2 & {K^T}_{xx} \\ {K^2}_{xx} & K_{xx'} \end{bmatrix}\right)$$

Thus, we are actually estimating **Posterior Distribution** f(x’) from **Prior Distribution** f(x), such that P(f(x’) | f(x)), which we can estimate by conditioning the Joint Distribution P(X, Y).



# Associative Model

So far, **Stochastic Causality** has been understood only from the **Perspective of Probability**, such that:

$$\text{If } P(\mathbf{X}), \text{ what is } P(\mathbf{Y}), \text{ given } \{\mathbf{S}_1, \ldots, \mathbf{S}_n\}$$

Now, we will approach **Stochastic Causality** as an **Associative Function**, such that:

$$P(Y \mid X) \sim F(X, Y)$$

Such Associative Function defines the **Score of the Association** between X and Y, given { $\mathbf{S}_1, \ldots, \mathbf{S}_n$ }.

One of the most famous Associative Approaches is the **Energy Based Model** (EBM), in which the score of dependency is defined as **Scalar Energy**.

Thus, in Associative Models, the dependency between X and Y is defined as a **Associative Affinity**, instead of **Stochastic Causality**.

# Energy Based Model

We define an energy function **F**: **X** × **Y** ⟶ R where F(x, y) describes the level of dependency between (**x**, **y**) pairs; in other words, it is to predict if a certain pair of (x, y) fits together or not.

$$y' = \text{argmin}_y \{ F(x, y) \}$$

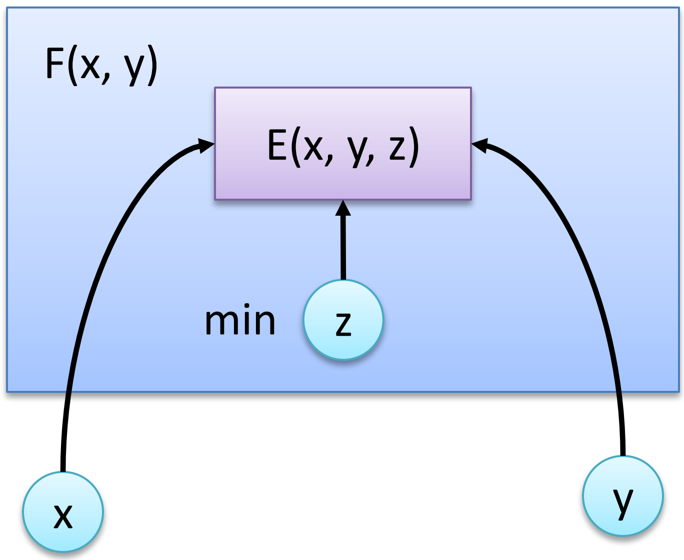
Actually, the output **y** depends on **x** as well as an extra variable **z**, known as the latent variable, which we do not know the value of. These latent variables can provide auxiliary information.

To do inference with latent variable EBM, we want to simultaneously minimize energy function with respect to y and z.

$$y', z' = \text{argmin}_{y,z} E(x, y, z)$$

Actually, latent variable **Z** represents the possible associative information between **X** and **Y**. Thus, we would want such associative information to be as minimal as possible.

# Energy Based Model




In light of Latent Variable **z**, the Associative Energy Function F(x, y) can be redefined:

$$F_\infty(x,y) = min_z E(x,y,z)$$

However, since the exact minimum value of z is not possible, we will have to marginalize over z:

$$F_\beta(x,y) = -\frac{1}{\beta}\log\int_z e^{-\beta E(x,y,z)}$$

$$y' = argmin_y F_\beta(x,y)$$

# Energy Based Model

Although Energy Function defines the **Associative Score**, we could always convert it back to **Probability**, by using Gibbs-Boltzmann Distribution.

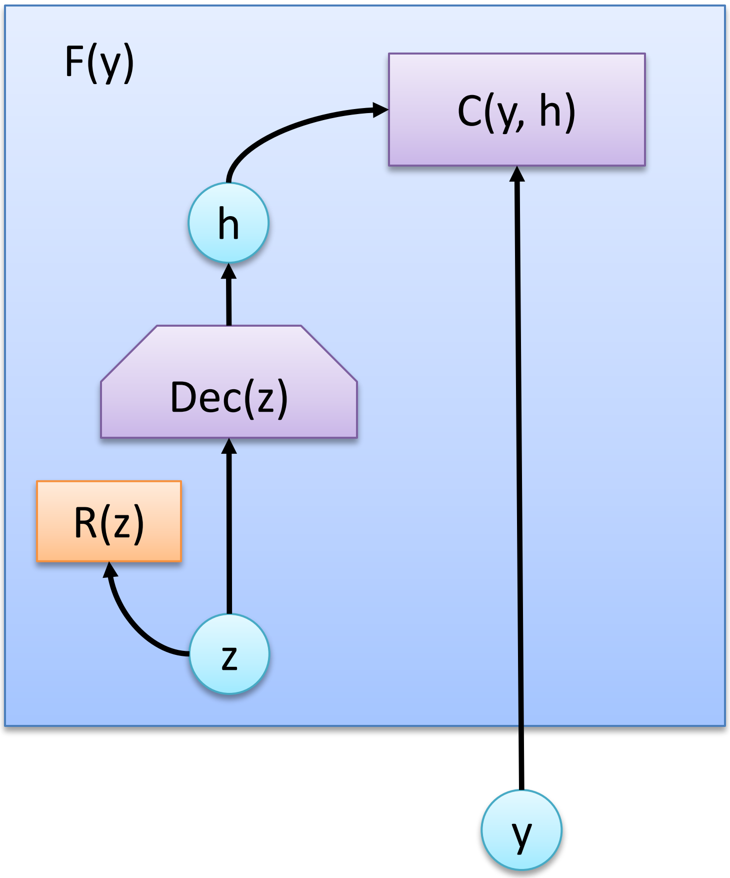
$$P(Y \mid X) = \frac{e^{-\beta F_{\beta}(x,y)}}{\int_{y'} e^{-\beta F_{\beta}(x,y')}}$$

However, it is not always desirable to be represent **Associative Score** as **Probability**.

Since, the marginalization of latent variable **z** defines the amount of information (energy) between x and y, it is also known as "**Free Energy**":

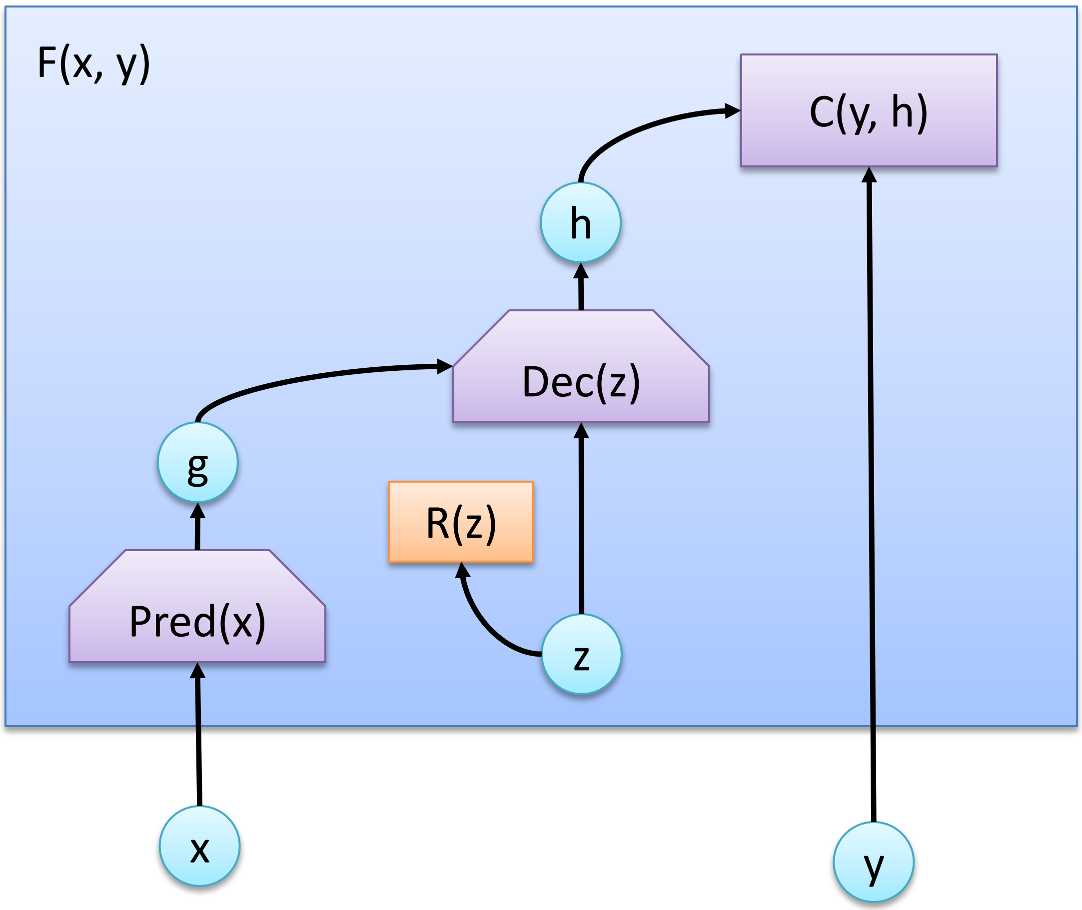
$$F_{\beta}(x,y) = -\frac{1}{\beta}\log\int_{z} e^{-\beta E(x,y,z)}$$

# Unconditional Energy Function




**Unconditional Energy Function** can define the Minimum Latent Information z for the Representation of Y, which is primarily used in **Representational Learning**.

# Conditional Energy Function




However, **Conditional Energy Function** can define the Minimum Associative Information between X and Y.

# Stochastic Learning

Now, let us look back into Stochastic Learning, after defining various Stochastic Reasoning Models. As we already know Stochastic Learning is to estimate possible **State Space**, such that:

If P(**X**), what is {$\mathbf{S}_1$, …, $\mathbf{S}_n$ }, given P(**Y**)

However, we can view **Stochastic Learning** as an **Optimization Process**, which is to **Minimize the Uncertainty** of Prediction or Observation.

# Maximum Likelihood

In **Maximum Likelihood Learning**, we attempt to find the **Parameters** (State Space) of a distribution that maximize the likelihood of observing the data.

If θ represents the parameters of a distribution, then the maximum likelihood estimate is:

$$\theta' = argmax_{\theta}\, P(D \mid \theta)$$

Since we can define **Observation of Data** as **(X, Y)**, which is Independently and Identically Distributed,

$$P(D \mid \theta) = \prod_{i} P(x_i, y_i \mid \theta)$$

Then, log-likelihood is:

$$\theta' = argmax_{\theta} \sum_{i} \log P(x_i, y_i \mid \theta)$$

# Maximum Likelihood

Since we are assuming Data is Independently and Identically Distributed, they are conditionally independent.

$$P(x_i, y_i \mid \theta) = P(x_i \mid \theta)P(y_i \mid x_i, \theta)$$

Since $P(x_i \mid \theta)$ is independent of the Parameters, and thus, it can be a constant.

$$P(x_i, y_i \mid \theta) = kP(y_i \mid x_i, \theta)$$

Thus, Optimization of $P(x_i, y_i \mid \theta)$ can be reduced to Optimization of $P(y_i \mid x_i, \theta)$, which can be denoted as:

$$P(x_i, y_i \mid \theta) \sim L(y_i \mid x_i, \theta)$$

Finally,

$$\theta' = argmax_\theta \sum_i \log L(y_i \mid x_i, \theta)$$

# Maximum Posteriori

Maximizing Posterior is actually Bayesian approach to Stochastic Learning, which involves estimating P(θ | D), the posterior distribution over θ given our data D :

$$\theta' = argmax_\theta \; P(\theta \mid D)$$

Thus, Maximizing Posterior requires to define the Prior Distribution of Model P(θ).

$$P(\theta \mid D) = \frac{P(D \mid \theta)P(\theta)}{P(D)}$$

Since, the Probability Distribution of Data P(D) is independent of the Distribution of Model, P(D) will be constant.

Thus, Maximizing Posterior is to maximize Likelihood and Prior.

$$P(\theta \mid D) \propto \; P(D \mid \theta)P(\theta)$$

$$\theta' = argmax_\theta \; P(D \mid \theta)P(\theta)$$

# Maximum Expectation

Since both **Maximum Likelihood** and **Maximum Posteriori** are **Point Estimate**, sometimes, it is necessary to define Maximum Expectation (Average) :

$$\theta' = \int_{\theta} \theta \, P(\theta \mid D) d\theta$$

Unlike **Maximum Likelihood** and **Maximum Posteriori**, Maximum Expectation is not point estimate, it is to estimate the average of a distribution.

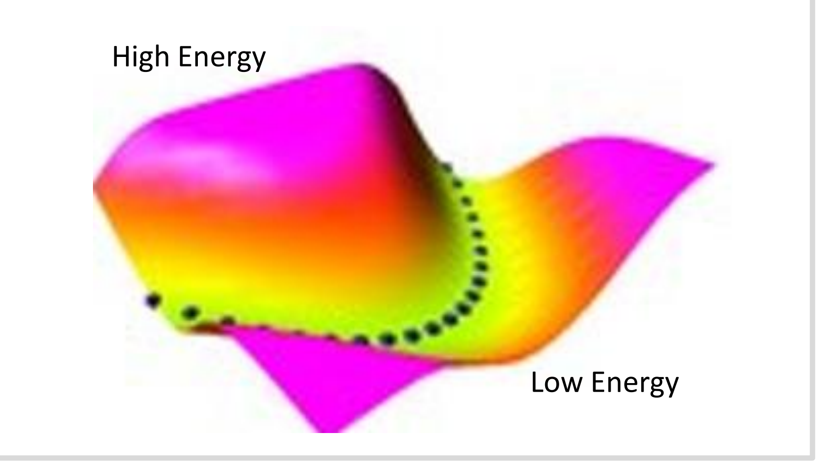
# Associative Learning

In view of Associative Models, we will look into **Associative Learning**, which is to maximize the Associative Score.

Actually, Associative Learning is more general than Stochastic Learning, as it is possible to define an associative score between X and Y more easily.

In addition, Associative Learning is more aligned with the actual learning in Biological Neural Networks, as Hebbian suggested, "neurons that fire together wire together".

In Energy Based Model, Associative Learning is to optimize the **Energy Level** of Association between X and Y, such that high energy level defines low associations and vice versa.



# Energy Optimization

Let us define an **Energy Function $f_\theta$** , a general form of Boltzmann Distribution, which is a subset of **Exponential Distribution**.

$$P_\theta(x) = \frac{1}{Z_\theta} e^{-f_\theta(x)}$$

If we take **Gradient** of log $P_\theta(x)$, we get :

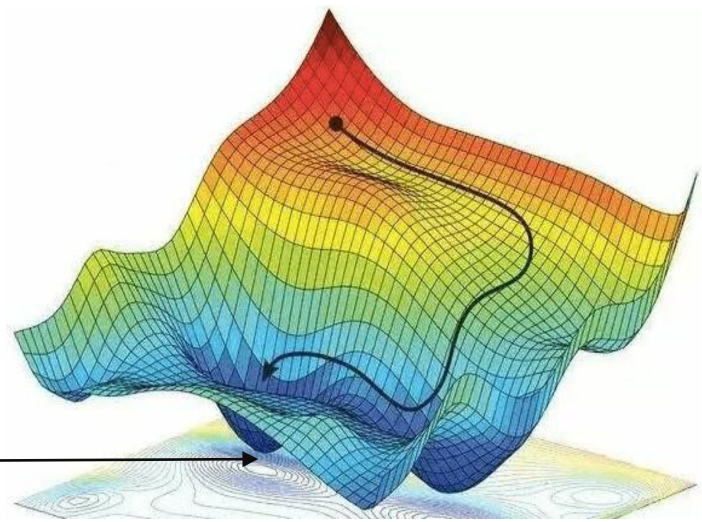
$$\begin{aligned}
\nabla_{\boldsymbol{x}} \log p_{\boldsymbol{\theta}}(\boldsymbol{x}) &= \nabla_{\boldsymbol{x}} \log(\frac{1}{Z_{\boldsymbol{\theta}}} e^{-f_{\boldsymbol{\theta}}(\boldsymbol{x})}) \\
&= \nabla_{\boldsymbol{x}} \log \frac{1}{Z_{\boldsymbol{\theta}}} + \nabla_{\boldsymbol{x}} \log e^{-f_{\boldsymbol{\theta}}(\boldsymbol{x})} \\
&= -\nabla_{\boldsymbol{x}} f_{\boldsymbol{\theta}}(\boldsymbol{x})
\end{aligned}$$

# Energy Landscape

As we can see, the Gradient of Energy Function (**$f_\theta$**) is:

$$\nabla_x f_\theta(x) = -\nabla_x log P_\theta(x)$$

Gradient of Energy Function defines the Gradient of Uncertainty (Negative Gradient of Probability) in Energy Landscape.



Minimum
Uncertainty

# Learning in EBM

In Energy Based Model, Associative Learning is to optimize the energy level of association between X and Y.

There are two primary methods, which are generally used to achieve this.

The first method is known as **Contrastive Method**, which is to increase the contrast between X and Y as much as possible. **Maximum Likelihood** method is also a contrastive method.

The other method is known as **Regularization Method**, which is to regulate the association between X and Y. **Variational Auto Encoder (VAE)** is known as a regularization method.

# Score Based Model

Actually, **Energy Based Model** (EBM) could be thought of as a special **Score Based Model**, and Score Based Model, in turn, could also be a special **Associative Model**.

Thus, we can consider Scored Based Model (SBM) as a more general approach than Energy Based Model (EBM), in which **Associative Dependency** is calculated as a **Scalar Score (Energy)**, instead of a **Probability**.

However, in Scored Based Model (SBM), the **Associative Score** does not always need to be a **Scalar**, which can even be a **Tensor**.

Thus, Scored Based Model (SBM) generally employs **Fisher Divergence**, which could be used as **Score Matching**, more often than **KL Divergence**.

# Score Based Model

In Scored Based Model (SBM), the **Gradient** of log $P_\Theta(x)$ is estimated with a Score Function $\mathbf{S_\theta}$, which can be either a Scalar or a Tensor function:

$$\nabla_x \log P(x) \sim S_\theta$$

Where

$$S_\theta = \nabla_x \log Q(x)$$

If the Score Function $\mathbf{S_\theta}$ is a Scalar Function, it could be represented as **Energy Function**. However, we will define Score Function $\mathbf{S_\theta}$ for a more generic **Tensor Function**.

# KL Divergence

**Kullback–Leibler** Divergence (KL Divergence) between two probability distributions P(x) and Q(x) can be defined as a **Relative Entropy** :

$$D_{KL}(P \,||\, Q) = \int P(x)\, log\left(\frac{P(x)}{Q(x)}\right) dx$$

KL Divergence is generally used in Stochastic Optimization as a **Regularization**.

# Fisher Divergence

Given a Score Function $\mathbf{S_\theta}$, Fisher Divergence, more precisely **Fisher Information Metric**, can be defined as:

$$F_\theta = E_p[||\nabla_x \log P(x) - S_\theta||^2]$$

Or

$$F_\theta = \int P(x)\,[||\nabla_x \log P(x) - S_\theta||^2]dx$$

More generally, Fisher Information Metric can be used to measured the **Statistical Manifold** in **Information Geometry**.

$$F = E_p[||\nabla_x \log P(x) - \nabla_x \log Q(x)||^2]$$

# Score Based Learning

One of the major advantages of Score Based Learning is it is possible to avoid the intractable **Normalization**, usually defined by **Partition Functions**, which thus simplifies the optimization.

In addition, Score Based Learning seems to be more aligned with Human's Evaluation, as we are more likely to give a **Score** than a **Probability**.

For example, we would rarely say what would be the probability of a candidate to get hired for a job. Instead, we would just assign a matching score between a candidate and the job.

Furthermore, a score can be a **Multivariate Tensor**, which could represent the Multidimensional Aspects of Evaluation.

# Score Based Generative Model

Give a score function $\mathbf{S}_\theta$, it is generally possible to define a Generative Model, by utilizing an interactive procedure known as **Langevin dynamics**.

Langevin Dynamics provides an MCMC (Monte Carlo Markov Chain) procedure to sample from a distribution using only its score function $\mathbf{S}_\theta$, such that:

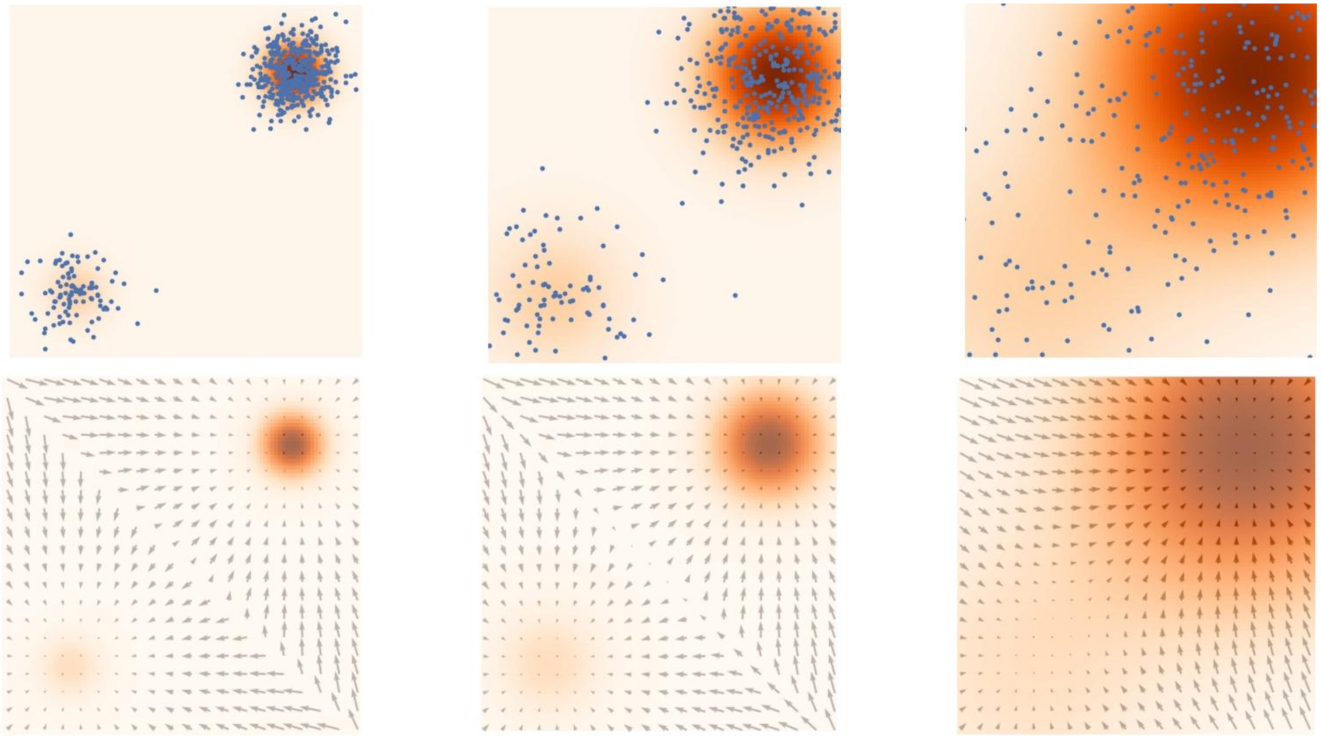
$$x_{i+1} \leftarrow x_i + \in S_\theta + \sqrt{2} \in z_i$$

where

$$z_i \sim \mathcal{N}(0, I)$$

# Scored Based Generative Model

Score Based Generative Model is also known as **Noise Conditional Score Network** (NCSN):



Source: https://yang-song.net/blog/2021/score/

# Score Based Policy Optimization

Policy Optimization, also known as Reinforced Learning, usually employs the **Policy Gradient**, which is defined as:

$$\nabla_\theta \log P(\tau \mid \theta) = \sum_{t=0}^{T} \nabla_\theta log\pi_\theta(a_t \mid s_t)$$

Thus, the Gradient of **Expected Return** becomes:

$$\nabla_\theta U(\pi_\theta) = \int P(\tau \mid \theta) \nabla_\theta \log P(\tau \mid \theta) R(\tau)$$

$$\nabla_\theta U(\pi_\theta) = \int P(\tau \mid \theta) \sum_{t=0}^{T} \nabla_\theta log\pi_\theta(a_t \mid s_t)\, R(\tau)$$

# Score Based Policy Optimization

The reward of a trajectory **R($\tau$)** can be reduced as a **Reward-to-go**, which is usually represented by **$Q^{\pi}(a_t, s_t)$**, by discarding the past rewards.

$$R(\tau) = \sum_{t'=t}^{T} Q_{\varphi}(s_{t'}, a_{t'})$$

By adding a **baseline**, which is usually represented by **$U^{\pi}(s_t)$**, we can define an **Advantage Function $A^{\pi}(a_t, s_t)$** such that:

$$A^{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - U^{\pi}(s_t)$$

Thus,

$$\nabla_{\theta} U(\pi_{\theta}) = \int P(\tau \mid \theta) \sum_{t=0}^{T} \nabla_{\theta} log \pi_{\theta}(a_t \mid s_t) \left( \sum_{t'=t}^{T} A_{\varphi}(s_{t'}, a_{t'}) \right)$$

# Scored Based Policy Optimization

**Policy Optimization** is usually a **Policy Iteration**, which alternates between **Policy Evaluation** — computing the value function for a policy — and **Policy Improvement** — using the value function to obtain a better policy.

In Actor Critic  Method, the **policy** is referred to as the **actor**, and the **value function** as the **critic**.

Many Policy Optimizations usually consider the **Entropy of the Policy**, but instead of maximizing the entropy, they use it as an **Regularizer**.

The Entropy of a Policy can be defined as:

$$H(\pi) = E_x[\pi(.|s_t)]$$

# Soft Actor Critic

In Soft Actor Critic (SAC), **Entropy of the Policy** is also optimized, along with the reward.

$$\nabla_\theta U(\pi_\theta) = \sum_{t=0}^{T} E_{(a_t,s_t)}[r(a_t,s_t) + \alpha H(\pi_\theta(.|s_t))]$$

where $\alpha$ is the temperature and the Entropy of a Policy $H(\pi_\theta)$ is:

$$H(\pi_\theta) = E_x[\pi_\theta(.|s_t)]$$

Thus, the agent gets a **bonus reward** at each time step proportional to the entropy of the policy.

Actually, we can consider such bonus reward as a **Score** to guide the policy optimization.

# Score Based Evaluation

One of the earliest Score Based Evaluations is **Sentiment Analysis**.

Given a sentence, what is the Score of the Sentiment, which is usually defined as a Scalar Score, from being **Negative** to **Positive**, ranging from $-\infty$ to $+\infty$.

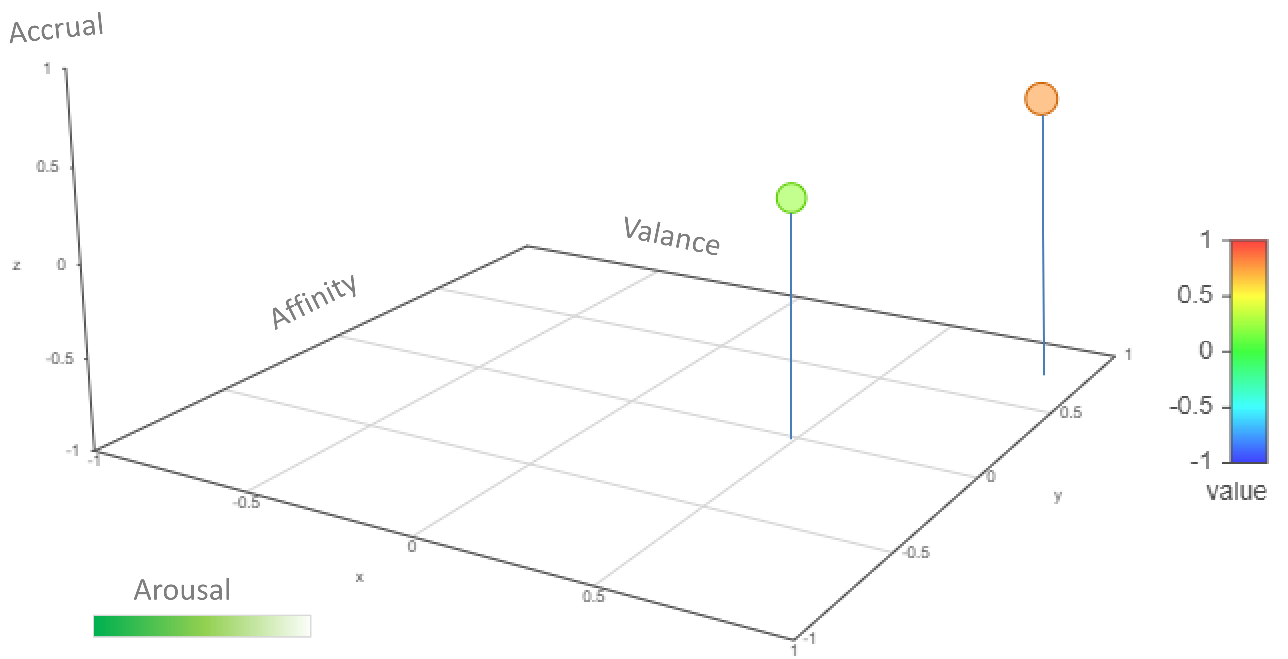
However, Score Evaluation can be extended for **Multidimensional Metric**, represented by a **Tensor**, instead of a **Scalar**.

Actually, in addition to text, we can compute a **Score** for any **Modality** of Data, without any reference to Externalities.

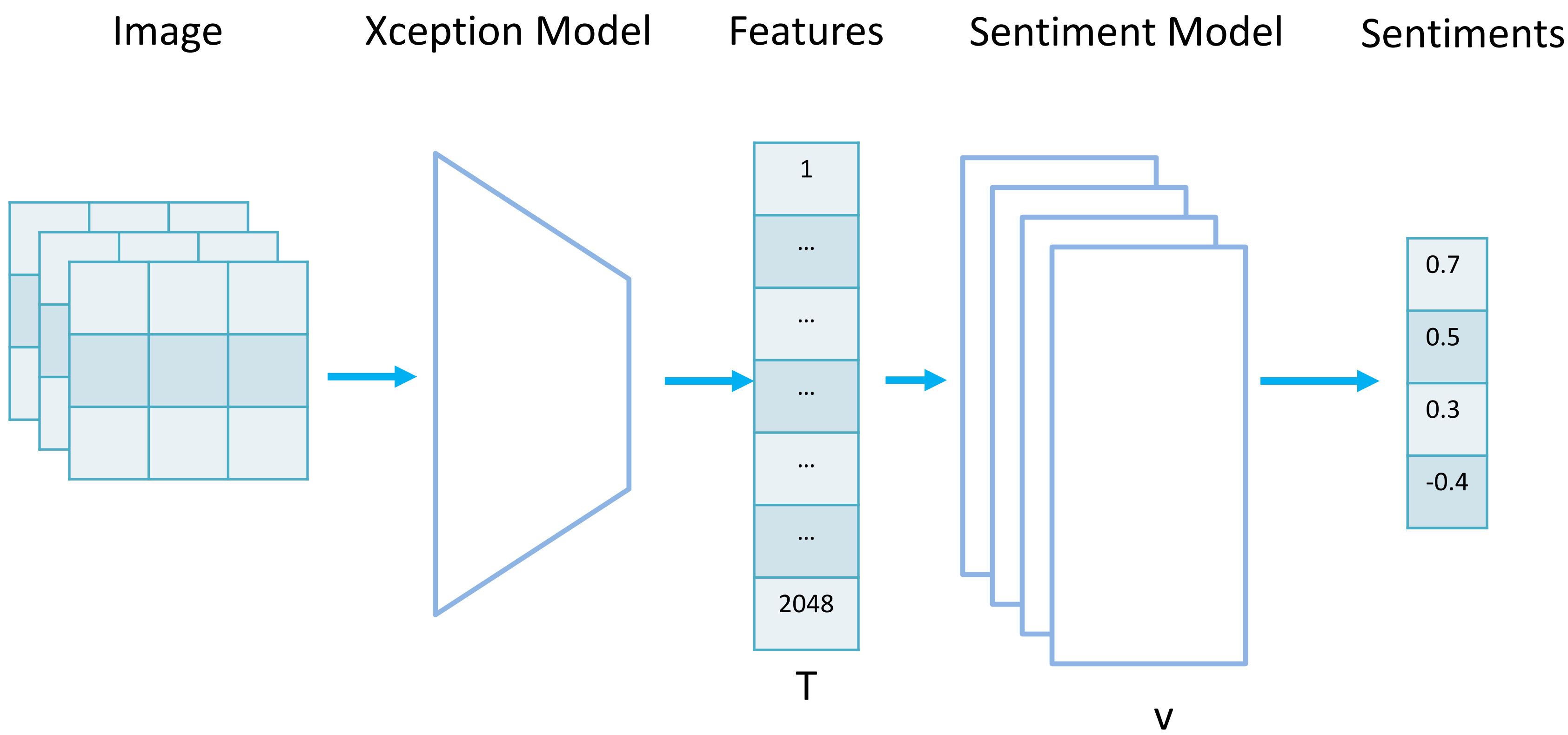
For example, we can measure the Sentiment Score of an Image in Multidimensional Metric, defined by Vector Sentiment Space.

# Sentiment Space

The sentiment space is defined as k = 4 Dimensional Vector Space, which is a Linear Space of  4 bases : { Valance, Arousal, Affinity and Accrual }.



$$L_v T = L(v, T)$$




Sentiment Evaluation is based on the concept that a Tensor **T** is evaluated in terms of a vector **v** in **Sentiment Space** (V3A), which would give us a **sentiment score**.

# Sentiment Scores

As we can see, each Dense Layer (MLP) can be considered as **sub manifold** of **the coordinate function** φ such that:

$$\varphi: M \rightarrow N \rightarrow R$$ , where M = 2048, N = 512, R = 1

$$x_j = \varphi_j(T)$$, where T = Input Tensor

$$\boldsymbol{v} = \sum_{j=1}^{k} x_j v_j = [x_1 v_1, \cdots, x_k v_k]$$ , where **v** = Output Tensor

Therefore, $$\boldsymbol{v} = \sum_{j=1}^{k} \varphi_j(T) v_j$$

As we can see, **v** is a vector in Vector Space with k-dimension. However, I will define such vector space as an Equivalent of Sentiment Space.

# Advantages of Score Based Model

Generally, Probabilistic Models are defined such that:

If P(**X**), what is P(**Y**), given {$\mathbf{S}_1$, …, $\mathbf{S}_n$ }

In Bayesian Inference, it is to compute the Probability of Y, given X and S:

$$P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S})$$

However, in Score Based Model (SBM), we will no longer compute probability; instead, P (Y | X, S) is directly estimated with a **Score Function** such that:

$$P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S}) \sim F(X, Y, S)$$

For example, instead of computing what is the Probability of a Candidate getting hired for a Job, we can directly give a Score matching between a Candidate and a Job.

# Advantages of Score Based Model

Of course, computing a Score is generally less accurate than computing a Probability.

However, most of the time, computing Probability is generally **intractable** as it requires to compute the **Population** from a **Sample Space**.

Thus, SBM would release us such a burden.

In addition, Score Based Model would allow us to compute a Score independently in parallel, from which we could compute the Average Score.

For example, multiple interviewers can give each independent score to a job applicant, which could be combined and readjusted later.

Since computing a Score is generally less accurate than computing a Probability, for a small dataset, SBM will be less accurate. However, with bigger dataset, the accuracy of SBM will be greatly improved.

# Score Based Learning

In my humble opinion, Score Base Learning is more aligned with how humans usually learn.

Given a sensational experience, humans rarely try to compute a Probability; instead, we would try to compute a Score, which we would normally interpret as an Emotion, which could possibly be a Biological Score.

What is the probability of becoming a friend with a random stranger, during the first meet-up?

We don’t know; or we might not even care about it.

However, we would be giving a score, even subconsciously, in multidimensional metric, in terms of personality, friendliness, looks, and so on, which we would interpret as “First Impression”.